Nevomelanocytic Neoplasms

Nevomelanocytic neoplasias are commonly subcategorized based on the location of nevus cells:
- **Junctional**: cells located in the epidermis
- **Intradermal**: cells located in the dermis
- **Compound**: cells present in both the epidermis and dermis

Pathologically, nevus cells in the deeper dermis tend to be smaller and show reduced expression of melanocytic antigens.

Clinical Features

- Lesions are typically uniform in appearance and small in size
- Primarily develop during childhood and young adulthood
- Increased numbers of acquired nevi are associated with an elevated risk for melanoma (see Chap. 124)

Synonyms

Commonly used terms include:
- Nevus cell nevus
- Nevocellular nevus
- Nevocytic nevus
- Soft nevus
- Neuronevus
- Pigmented nevus
- Pigmented mole
- Common mole
- Melanocytic nevus
- Hairy nevus
- Cellular nevus
- Benign melanocytoma

Epidemiology

The number of nevi varies with age and sex:
- **First decade**: median of 0 (males), 2 (females)
- **Second decade**: 10 (males), 16 (females)
- **Third decade**: 16 (males), 24 (females)
- **Fourth decade**: 10 (males), 19 (females)
- **Fifth decade**: 12 (males), 15 (females)
- **Sixth decade**: 4 (males), 12 (females)
- **Seventh to ninth decades**: 2.0 (males), 3.5 (females)

In a study of Australian whites:
- Average nevus count peaked at **43 in males** and **27 in females** during the second and third decades, respectively
- Numbers declined significantly by the sixth and seventh decades

Similar age-related patterns have been observed globally. Gender differences in nevus prevalence are not consistently evident, though most studies report nearly equal distribution between males and females.

Ethnic and Phenotypic Variations

- Prevalence of acquired nevi varies by ethnicity
- In African populations, individuals with lighter skin have a higher nevus count than those with darker skin
- In prepubertal white children, higher nevus counts are associated with:
- Pale skin
- Blue or green eyes
- Blond or light-brown hair
- Tendency to sunburn
- *No significant association with freckling*
- Other studies report variable associations with these phenotypic traits

Environmental Factors

- **Ultraviolet radiation (UVR)** is a key contributing factor in nevus development
- In Australia, nevus density increases with higher ambient sun intensity in northern regions
- Use of UVR-blocking sunscreens in children has been shown to reduce the formation of new nevi

Genetic Influences

- Family aggregation of nevus size, number, and distribution is well documented
- Strong hereditary component, particularly evident in **atypical nevi** and **familial melanoma** syndromes
- Also relevant in **congenital melanocytic nevi (CNN)**

Immune-Mediated Phenomena

- Acquired nevi can be targeted by the immune system, leading to **halo nevus** formation
- In a study of 8,298 white adolescents (12–16 years):
- **1.2% of males** (51 individuals)
- **0.5% of females** (22 individuals)
- Overall prevalence: **0.9%**
- In a large series of halo nevi:
- Age range: 3 to 42 years
- Mean age: 15 years
- Most cases occur before age 20

Etiology and Pathogenesis

The exact origin of nevi remains unclear, despite numerous theories and some supporting evidence.

Current Understanding:

- Nevomelanocytic nevi are **clonal proliferations**
- A **B-RAF V600E mutation** is identified in the majority of:
- Common acquired nevi
- Atypical (dysplastic) nevi
- Congenital nevi
- This mutation is thought to play a central role in nevus initiation

Exceptions:

- **B-RAF mutations are not commonly found** in:
- Blue nevi
- Spitz nevi

Thus, while B-RAF activation is a key driver in most melanocytic nevi, alternative molecular pathways likely underlie other nevus subtypes.